## ANSWERS TO CASE SCENARIO 1

| Question No. | Answer |
|---|---|
| 1. | **(c)** Stock A = 0.3 x 16 + 0.4 x 14 + 0.3 x (-9) = 7.70<br>Stock B = 0.3 x 19 + 0.4 x 16 + 0.3 x (-7) = 10 |
| 2. | **(d)** $(18 - 9)^2 0.3 + (15 - 9)^2 0.4 + (-8 - 9)^2 0.3 = 125.4$ |
| 3. | **(c)** Stock A = 122.70/ 125.40 = 0.9785<br>Stock B = 125.40/ 125.40 = 1 |
| 4. | **(a)** Stock A = 8% + 0.9785(9% - 8%) = 8.9785%<br>Stock B = 8% + 1(9% - 8%) = 9% |
| 5. | **(c)** While Stock A's expected rate of return is lower than that required return. However, for Stock B the reverse is true. |

## ANSWERS TO CASE SCENARIO 2

| Question No. | Answer |
|---|---|
| 1. | **(b)** 20000 x 25 + 30000 x 350 + 38000 x 290 + 50000 x 400 |
| 2. | **(c)** ₹ 420.20 lakhs - ₹ 4.50 lakhs = ₹ 415.70 lakhs |
| 3. | **(c)** ₹ 415.70 lakhs/ 20 lakhs = ₹ 20.785 |

## ANSWERS TO CASE SCENARIO 3

| Question No. | Answer |
|---|---|
| 1. | **(b)** (₹ 2160000 x 1.3)/ (2310 x ₹ 240) = 5.06 rounded off to 5 |
| 2. | **(d)** 9000(₹ 212 – ₹ 240) + (2310 – 2025) x 5 x ₹ 240 = ₹ 90,000 |
| 3. | **(b)** 9000(₹ 255 – ₹ 240) + (2310 – 2385) x 5 x ₹ 240 = ₹ 45,000 |

## ANSWERS TO CASE SCENARIO 4

| Question No. | Answer | |
|---|---|---|
| 1. | (c) | [(£ 0.81 x 0.30 + £ 0.81 x 0.50 + £ 0.83 x 0.20) x 20000000] - £ 400000 |
| 2. | (d) | £ 0.791 x $ 20000000 |
| 3. | (a) | (£ 0.78 x 0.30 + £ 0.81 x 0.50 + £ 0.83 x 0.20) = £ 0.805; £ 16.10 million |
| 4. | (d) | As no hedging strategy has highest receipt of £. |

## ANSWERS TO CASE SCENARIO 5

| Question No. | Answer | |
|---|---|---|
| 1. | (c) | [(90420 – 42420)/90420] x 100 = 53.09% |
| 2. | (c) | [(90420 – 33378)/90420] x 100 = 63.09% |
| 3. | (c) | [(90420 – 78540)]/90420 x 100 = 13.14% |

## ANSWERS TO CASE SCENARIO 6

| Question No. | Answer | |
|---|---|---|
| 1. | (d) | ₹ 1,00,000 + ₹ 6,00,000 + ₹ 28,000 + ₹ 22,500 + ₹ 25,000 - ₹ 2,50,000 |
| 2. | (c) | 175% x ₹ 25,00,000 + ₹ 4,00,000 + ₹ 2,50,000 + 90% x ₹ 2,50,000 + ₹ 5,25,500 |
| 3. | (b) | (₹ 57,75,500 – 5,00,000 x ₹ 0.60)/ 5,00,000 |

## ANSWERS TO CASE SCENARIO 7

| Question No. | Answer | |
|---|---|---|
| 1. | (a) | ₹ 25 x 100 + ₹ 8 x 100 |

| | | |
|---|---|---|
| **2.** | **(c)** | 100 (₹ 600 - ₹ 500) - ₹ 3,300 |
| **3.** | **(b)** | 100 (₹ 820 - ₹ 750) - ₹ 3,300 |

## ANSWERS TO CASE SCENARIO 8

| Question No. | Answer | |
|---|---|---|
| **1.** | **(b)** | 60 + (150 - 60)15/30; 80 + (178 - 80) 15/30 |
| **2.** | **(c)** | ₹ 84.2525 + ₹ 0.0105; ₹ 85.5945 + ₹ 0.0129 |
| **3.** | **(d)** | $\frac{0.0105}{84.25775} \times \frac{12}{2.5}; \frac{0.0129}{85.60095} \times \frac{12}{2.5}$ |

## ANSWERS TO CASE SCENARIO 9

| Question No. | Answer | |
|---|---|---|
| **1.** | **(a)** | 16/20; 20/10 |
| **2.** | **(c)** | [(0.8 x 1200000) + (2.0 x 400000)]/[1200000 + (400000 x 20/16)] |
| **3.** | **(b)** | ₹ 16/ ₹ 1.04 |

## ANSWERS TO CASE SCENARIO 10

| Question No. | Answer | |
|---|---|---|
| **1.** | **(b)** | (₹ 1,200 - ₹ 1,100) x 200 + ₹ 70 x 200 - ₹ 50 x200 = ₹ 24,000 |
| **2.** | **(a)** | ₹ 1,200 – (₹ 70 - ₹ 50) = ₹ 1,180 |
| **3.** | **(a)** | (₹ 1,500 - ₹ 1,600) x 300 + ₹ 70 x 300 - ₹ 50 x 300 = - ₹ 24,000 |

## ANSWERS TO CASE SCENARIO 11

| Question No. | Answer | |
|---|---|---|
| **1.** | **(b)** | 720000 x 0.50 = 360000 |

| | | |
|---|---|---|
| **2.** | **(c)** | (₹ 72,00,000 + ₹ 14,40,000)/(2400000 + 360000) = ₹ 3.13 |
| **3.** | **(c)** | ₹ 3.13 x 0.50 = ₹ 1.57 |
| **4.** | **(c)** | (₹ 3.13 x 5 = ₹ 15.65) x 2760000 |

## ANSWERS TO CASE SCENARIO 12

| Question No. | Answer | |
|---|---|---|
| **1.** | **(b)** | 8.45% x 0.50 + 22% x 0.50 = 15.23% |
| **2.** | **(a)** | 1000000(1 – 0.35) – [{10.40% x 0.80 + 26% x 0.20} 4000000] |
| **3.** | **(a)** | [{1000000(1 – 0.35) – 0.20 x 4000000 x 9.75%} x 10]/200000 |
| **4.** | **(b)** | [{1000000 x (1 – 0.35) – 0.50 x 4000000 x 8.45%} x 10] |
| **5.** | **(a)** | [{1000000 x (1 – 0.35) – 0.80 x 4000000 x 10.40%} ]/ 122000 |

## ANSWERS TO CASE SCENARIO 13

| Question No. | Answer | |
|---|---|---|
| **1.** | **(a)** | Conversion value = Market price of share × No. of shares per bond = ₹ 48 × 20 = ₹ 960 |
| **2.** | **(b)** | Conversion premium = Market price of convertible bond – Conversion value = ₹ 1060 – ₹ 960 = ₹ 100 |
| **3.** | **(c)** | Straight value represents the downside protection or investment value for the investor. |
| **4.** | **(c)** | The upside potential arises from the appreciation in the underlying equity share price. |
| **5.** | **(b)** | $\left(\frac{1060}{20}-48\right)\Big/48$ |

## ANSWERS TO CASE SCENARIO 14

| Question No. | Answer | |
|---|---|---|
| 1. | **(b)** | β = Correlation × (σ share / σ market)<br>= 0.3733 × (18% / 12%) = 0.3733 × 1.5 ≈ 0.56 |
| 2. | **(c)** | Equity risk premium = Market return − Risk-free rate<br>= 20% − 13% = 7% |
| 3. | **(c)** | Different growth rates are applicable for two different phases. |
| 4. | **(b)** | Very often the price of over-valued share is driven by hype, speculation, or temporary optimism rather than solid fundamentals. |

## ANSWERS TO CASE SCENARIO 15

| Question No. | Answer | |
|---|---|---|
| 1. | **(b)** | Since Coupon rate (6%) < YTM (10%), hence bond trades at a discount. |
| 2. | **(b)** | It can be understood as the average time taken to receive the cash flows on a given bond. |
| 3. | **(a)** | Bond X has a longer maturity and higher duration, hence higher interest rate risk. |
| 4. | **(b)** | Bond Y has shorter duration hence lower interest rate risk. |

## ANSWERS TO CASE SCENARIO 16

| Question No. | Answer | |
|---|---|---|
| 1. | **(c)** | When an option lapses, the maximum loss equals the premium paid. |
| 2. | **(d)** | Spot price (160) > Exercise price (140), hence a call option is in-the-money. Spot price (175) < Exercise price (200), hence a call option is in-the-money. |
| 3. | **(d)** | Intrinsic value of put option = Exercise price − Spot price<br>= ₹ 4000 – ₹ 3500 = ₹ 500 |
| 4. | **(a)** | In Situation I, intrinsic value equals premium, resulting in zero profit/loss. |
| 5. | **(c)** | Spot price is less than Exercise Price and hence Option is exercised. |

## ANSWERS TO CASE SCENARIO 17

| Question No. | Answer | |
|---|---|---|
| 1. | **(a)** | ₹ 315 Lakh x 10.00 = ₹ 3150 Lakh |
| 2. | **(b)** | This will increase the outflow of cash on account of higher interest as well as will increase in financial risk. |
| 3. | **(b)** | Post acquisition the EPS will be reduced to ₹ 58.37. |

## ANSWERS TO CASE SCENARIO 18

| Question No. | Answer | |
|---|---|---|
| 1. | **(c)** | A future borrower buys FRA at the offer rate to lock in borrowing cost. |

| | | |
|---|---|---|
| **2.** | **(c)** | FRA compensates the borrower when market rates exceed the agreed rate. |
| **3.** | **(b)** | FRA is settled at the start of the underlying loan period. |
| **4.** | **(b)** | Settlement relates to a future interest period but is paid upfront. |
| **5.** | **(c)** | Except (c) other points features are related to FRA. |

## ANSWERS TO CASE SCENARIO 19

| Question No. | Answer | |
|---|---|---|
| **1.** | **(b)** | Future payment in pounds means exposure to appreciation of £. |
| **2.** | **(c)** | Forward contract locks in the exchange rate. |
| **3.** | **(c)** | Options provide downside protection with upside benefit. |
| **4.** | **(b)** | Market rate is better than exercise price. |
| **5.** | **(c)** | In remaining methods only one transaction in two currencies is involved. |

## ANSWERS TO CASE SCENARIO 20

| Question No. | Answer | |
|---|---|---|
| **1.** | **(a)** | Volatility is measured by standard deviation; X Ltd. has higher SD (40%). |
| **2.** | **(c)** | A correlation of 0.72 indicates strong positive relationship. |
| **3.** | **(b)** | Lower correlation improves diversification benefits. |
| **4.** | **(c)** | Other types of risks are systematic risks which cannot be reduced or eliminated through diversification. |
| **5.** | **(c)** | CAPM depends on beta, not total volatility. |

## ANSWERS TO CASE SCENARIO 21

| Question No. | Answer |
|---|---|
| 1. | **(a)** Cash flows and discount rate must be consistent. |
| 2. | **(b)** Ignoring inflation understates future cash flows. |
| 3. | **(c)** Depreciation is based on historical cost. |

## ANSWERS TO CASE SCENARIO 22

| Question No. | Answer |
|---|---|
| 1. | **(a)** ₹ 50 x 100 – [₹ 35 x 100 + ₹ 25 x100] = - ₹ 1,000 |
| 2. | **(a)** ₹ 600 – [₹ 35 + ₹ 25] = ₹ 540 |
| 3. | **(d)** Refer difference between Financial and Real Options. |

## ANSWERS TO CASE SCENARIO 23

| Question No. | Answer |
|---|---|
| 1. | **(a)** ₹ 20 x 25 = ₹ 500 |
| 2. | **(b)** $\frac{550-400}{550}\times 100$ |
| 3. | **(a)** $\frac{550-500}{500}\times 100$ |
| 4. | **(c)** $\frac{550}{25}$ = ₹ 25 |

## ANSWERS TO CASE SCENARIO 24

| Question No. | Answer | |
|---|---|---|
| 1. | (b) | $\sqrt{4.40} = 2.0976$ |
| 2. | (d) | 12.00 x 0.60 + 11.00 x 40 = 11.60 |
| 3. | (b) | Based on Coefficient of Variation |

## ANSWERS TO CASE SCENARIO 25

| Question No. | Answer | |
|---|---|---|
| 1. | (b) | 6 : 5 x 0.50 = 0.60<br>4 : 5 x 0.25 = 0.20<br>1 : 2 x 0.25 = 0.125<br>0.925 |
| 2. | (a) | 0.925 x 50000 = 46,250 |
| 3. | (c) | (1000000 + 300000) / (200000 + 46250) = 5.28 |
| 4. | (b) | Ratio of EPS |

## ANSWERS TO CASE SCENARIO 26

| Question No. | Answer | |
|---|---|---|
| 1. | (c) | $\frac{800000}{10.10} \times 10 = 792079.21$ |
| 2. | (a) | $\frac{400000}{10.30}(10.35-10.30)+9000 = 10941.75$ |
| 3. | (c) | $\frac{10941.75}{400000} \times \frac{365}{D} = 0.0966$ |

| | | |
|---|---|---|
| **4.** | **(a)** | $\dfrac{\dfrac{800000}{10.10}(10.00\text{-}10.10)}{800000} \times \dfrac{365}{D} = -11.66\%$ |

## ANSWERS TO CASE SCENARIO 27

| Question No. | Answer | |
|---|---|---|
| **1.** | **(d)** | $\dfrac{265-235}{235} \times 100 = 12.77\%$ |
| **2.** | **(b)** | $\dfrac{265-240}{240} \times 100 = 10.42\%$ |
| **3.** | **(c)** | ₹ 265/20 = ₹ 13.25 |
| **4.** | **(b)** | ₹ 30(3.352) + ₹ 250(0.497) = ₹ 224.81 |

## ANSWERS TO CASE SCENARIO 28

| Question No. | Answer | |
|---|---|---|
| **1.** | **(b)** | Client A is looking at Credit Rating. |
| **2.** | **(a)** | Holding security till maturity. |
| **3.** | **(b)** | As market will react lately. |

## ANSWERS TO CASE SCENARIO 29

| Question No. | Answer | |
|---|---|---|
| **1.** | **(b)** | 360000 x 0.50 = 180000 |
| **2.** | **(c)** | (₹ 36,00,000 + ₹ 7,20,000)/(1200000 + 180000) = ₹ 3.13 |
| **3.** | **(c)** | ₹ 3.13 x 0.50 = ₹ 1.57 |
| **4.** | **(c)** | ₹ 3.13 x 10 = ₹ 31.30 |

## ANSWERS TO CASE SCENARIO 30

| Question No. | Answer | |
|---|---|---|
| **1.** | **(a)** | 16%(1 – 0.35) x 0.80 + 26% x 0.20 = 13.52% |
| **2.** | **(b)** | 500000(1 – 0.35) – [{13%(1 – 0.35) x 0.50 + 22% x 0.50} 2000000] = 20500 |
| **3.** | **(c)** | [(500000 – 0.20 x 2000000 x 15%) (1 – 0.35) x 11]/100000 = ₹ 31.46 |
| **4.** | **(c)** | [(500000 – 0.80 x 2000000 x 16%) (1 – 0.35) x 11] = 1744600 |
| **5.** | **(b)** | [(500000 – 0.50 x 2000000 x 13%) (1 – 0.35)]/ 83000 = ₹ 2.90 |

## ANSWERS TO CASE SCENARIO 31

| Question No. | Answer | | |
|---|---|---|---|
| **1.** | **(d)** | Refer the concept of Early Delivery under Fate of Forward Contract. | |
| **2.** | **(c)** | Refer the concept of Early Delivery under Fate of Forward Contract. | |
| **3.** | **(b)** | Bank Sells at Spot Rate on 28 November 2023 | ₹ 85.22 |
| | | Bank Buys at Forward Rate of 31 December 2023 (85.27 + 0.15) | ₹ 85.42 |
| | | Swap Loss per US$ | ₹ 00.20 |
| | | Swap loss for US$ 1,00,000 | ₹ 20,000 |
| **4.** | **(b)** | On 28th November Bank sells at | ₹ 85.22 |
| | | It buys from customer at | ₹ 00.18 |
| | | Interest on Outlay fund for US$ 1,00,000 for 31 days | ₹ 00.20 |
| | | (US$100000 x 00.18 x 31/365 x 12%) | ₹ 183 |

| | | | |
|---|---|---|---|
| **5.** | **(c)** | Amount received on sale (₹ 85.40 x 1,00,000)<br>Less: Charges for early delivery payable to bank | ₹ 85,40,000<br>(₹ 20,183)<br>₹ 85,19,817 |

## ANSWERS TO CASE SCENARIO 32

| Question No. | Answer | |
|---|---|---|
| **1.** | **(b)** | First settle mutual obligation before proceeding further. |
| **2.** | **(d)** | Netting Schedule (US $ Thousands) |
| **3.** | **(b)** | |
| **4.** | **(b)** | |
| **5.** | **(a)** | (1560 – 200) x 0.01% x 1000 = 136 |

Netting Schedule (US $ Thousands)

| | **Receipt** | **Payment** | **Net Receipt** | **Net Payments** |
|---|---|---|---|---|
| France | 400 | 480 | --- | 80 |
| Germany | 360 | 260 | 100 | --- |
| UK | 420 | 320 | 100 | --- |
| Italy | 380 | 500 | --- | 120 |
| | | | **200** | **200** |

## ANSWERS TO CASE SCENARIO 33

| Question No. | Answer | |
|---|---|---|
| **1.** | **(b)** | $\frac{1000000-97550}{97550} \times \frac{12}{3} = 10.05\%$ |

| | | |
|---|---|---|
| **2.** | **(a)** | $\left[1+\frac{0.1005}{4}\right]^4 - 1 = 10.44\%$ |
| **3.** | **(c)** | 10.44% + 0.15 x 4 + 0.50 + 0.175 x 4 = 12.24% |
| **4.** | **(b)** | Issued by highly rated corporate entities to meet the short-term fund requirements. |
| **5.** | **(c)** | Refer the basic features of Commercial Papers. |

## ANSWERS TO CASE SCENARIO 34

| **Question No.** | **Answer** | |
|---|---|---|
| **1.** | **(b)** | Refer the concept of Pay Through Security. |
| **2.** | **(b)** | MAC bought receivables. |
| **3.** | **(c)** | Only for the purpose of issuing securities. |
| **4.** | **(d)** | Collects the payments from the receivables and pass to the SPV. |
| **5.** | **(b)** | Refer the concept of Stripped Securities. |

## ANSWERS TO CASE SCENARIO 35

| **Question No.** | **Answer** | |
|---|---|---|
| **1.** | **(a)** | Refer topic Empirical Evidence on Weak Form of Efficient Market Theory. |
| **2.** | **(c)** | Refer topic Empirical Evidence on Weak Form of Efficient Market Theory. |
| **3.** | **(b)** | Refer topic Empirical Evidence on Weak Form of Efficient Market Theory. |
| **4.** | **(b)** | Refer Filter Rules Test of Empirical Evidence on Weak Form of Efficient Market Theory. |
| **5.** | **(c)** | Refer conclusion of Case Scenario. |

## ANSWERS TO CASE SCENARIO 36

| Question No. | Answer | |
|---|---|---|
| 1. | (d) | Refer the concept of Repo and Reverse Repo. |
| 2. | (b) | $₹\ 9{,}942 + ₹\ 10{,}000 \times \frac{10}{100} \times \frac{262}{360} = ₹\ 10{,}670$ |
| 3. | (b) | $₹\ 8{,}00{,}00{,}000 \times \frac{10670}{10000} \times \frac{100\text{-}2}{100} = ₹\ 8{,}36{,}52{,}800$ |
| 4. | (a) | $₹\ 8{,}36{,}52{,}800 \times \left[1 + \frac{5.65}{100} \times \frac{14}{360}\right] = ₹\ 8{,}38{,}36{,}604$ |
| 5. | (a) | $₹\ 10{,}000 \times \frac{10}{100} \times \frac{262}{360} = ₹\ 728$ |

## ANSWERS TO CASE SCENARIO 37

| Question No. | Answer | |
|---|---|---|
| 1. | (b) | $\frac{18\%-9\%}{25\%-7\%} = 0.50$ |
| 2. | (c) | First compute Risk Free Rate of Return using simultaneous equations. Then use CAPM and compute the Expected Return under given scenario and multiply each return with 0.50 and add them. |
| 3. | (c) | 18% = α + 0.50% 25% or 9% = α + 0.50% 7% |
| 4.. | (d) | Refer the concept of Modern Portfolio Theory |
| 5 | (c) | Refer the concept of Capital Market Theory. |

## ANSWERS TO CASE SCENARIO 38

| Question No. | Answer | |
|---|---|---|
| 1. | (b) | Yield for 9 months = (153.33 x 9/12) = 115% |

| | | |
|---|---|---|
| | | Market value of Investments as on 31.03.2022<br>= 1,00,000/- + (1,00,000x 115%) = ₹2,15,000/-<br>Therefore, NAV as on 31.03.2022<br>= (2,15,000-10,000)/10,000 = ₹ 20.50 |
| **2.** | **(a)** | Since dividend was reinvested by Mr. X, additional units acquired $= \frac{10,000}{20.50}$ = 487.80 units<br>Therefore, units as on 31.03.2008 = 10, 000+ 487.80<br>= 10,487.80 |
| **3.** | **(c)** | Dividend as on 31.03.2023 = 10,487.80 x ₹ 10 x 0.2 = ₹ 20,975.60 |
| **4.** | **(c)** | Let X be the NAV on 31.03.2023, then number of new units reinvested will be ₹ 20,975.60/X. Accordingly 11296.11 units shall consist of reinvested units and 10487.80 (as on 31.03.2008). Thus, by way of equation it can be shown as follows:<br>$11296.11 = \frac{20975.60}{X} + 10487.80$<br>Therefore, NAV as on 31.03.2023=20,975.60/(11,296.11-10,487.80) = ₹ 25.95 |
| **5.** | **(c)** | NAV as on 31.03.2024 = ₹ 1,00,000 (1+0.7352x33/12)/ 11296.11 = ₹ 26.75 |

## ANSWERS TO CASE SCENARIO 39

| Question No. | Answer | |
|---|---|---|
| **1.** | **(b)** | Let N be the opening NAV, then<br>N (1 + 0.25) = ₹ 24.95<br>N = ₹ 19.96<br>i.e., beginning NAV = ₹ 19.96 |

| | | |
|---|---|---|
| **2.** | **(d)** | Let X be the number of units purchased<br>Then ending units = 20,800<br>Accordingly,<br>$20800 = X + \frac{0.998X}{24.95}$<br>$20800 = \frac{24.05X + 0.998X}{24.95}$<br>X = 20000 |
| **3.** | **(a)** | Initial NAV ₹ 19.96<br>Entry Load ₹ 0.04<br>₹ 20.00<br>Number of funds purchased 20,000<br>Amount of investment ₹ 4,00,000 |
| **4.** | **(b)** | Refer the concept of Expense Ratio in Mutual Funds. |
| **5.** | **(d)** | Refer the topic financial measures to evaluate Mutual Funds. |

## ANSWERS TO CASE SCENARIO 40

| Question No. | Answer | |
|---|---|---|
| **1.** | **(b)** | ₹ 120 x 0.05 + ₹ 140 x 0.20 + ₹ 160 x 0.50 + ₹ 180 x 0.10 + ₹ 190 x 0.15 = ₹ 160.50 |
| **2.** | **(a)** | Share can either be purchased from market or from the seller of the option. |
| **3.** | **(d)** | ₹ 10 x 0.50 + ₹ 30 x 0.10 + ₹ 40 x 0.15 = ₹ 14 |
| **4.** | **(b)** | When spot price exceeds exercise price it will be beneficial to purchase share from the option seller. |
| **5.** | **(d)** | In none of the case the expected spot price is ₹ 150. |

## ANSWERS TO CASE SCENARIO 41

| Question No. | Answer | |
|---|---|---|
| **1.** | **(b)** | Refer the concept of Credit Default Swap. |
| **2.** | **(c)** | CDS Contracts can also be used for speculation purpose as well. Regulated by ISDA. CDS provide protection against default risk only. |
| **3.** | **(b)** | Being OTC in nature. |
| **4.** | **(c)** | ₹ 500 crore x 1% x 3 = ₹ 15 crore |
| **5.** | **(c)** | ₹ 500 crore(1 – 0.75) = ₹ 125 crore |

## ANSWERS TO CASE SCENARIO 42

| Question No. | Answer | |
|---|---|---|
| **1.** | **(c)** | Other options are related to Profitability and Investment. |
| **2.** | **(b)** | Let the sale price/Unit be S so that the project would break even with 0 NPV. <br> $\therefore\ 10,00,000 = \frac{20,000 \times (S-40)}{1.1} + \frac{30,000 \times (S-40)}{1.21} + \frac{30,000\ (S-40)}{1.331}$ <br> S = ₹ 55.26 |
| **3.** | **(d)** | If sales price = ₹ 60 the cost price required to give a margin of ₹ 15.26 is (₹ 60 – ₹ 15.26) or ₹ 44.74. |
| **4.** | **(b)** | The requisite percentage fall is:- <br> 3,10,293/13,10,293 × 100 = 23.68% |
| **5.** | **(c)** | Since PV of inflows remains at ₹ 13,10,293 the initial outlay must also be the same. <br> Percentage rise = (3,10,293/10,00,000) x 100 = 31.03%. |

## ANSWERS TO CASE SCENARIO 43

| Question No. | Answer |
|---|---|
| 1. | **(b)** Long term bonds carry higher interest rate risk. |
| 2. | **(c)** 7.00% - 3.00% = 4.00% |
| 3. | **(c)** Investors are pessimistic about long term perspective of the economy. |
| 4. | **(b)** $[(1.06)^5/(1.05)^3]^{½} - 1 = 7.52\%$ |
| 5. | **(a)** $[(1.064)^7/(1.06)^5]^{½} - 1 = 7.41\%$ |

## ANSWERS TO CASE SCENARIO 44

| Question No. | Answer |
|---|---|
| 1. | **(b)** $\left[\frac{175\times8\%-125\times8\%}{50\times10\%}\right]$ = 0.8 year i.e. 9.6 months prior to 31.03.2023 |
| 2. | **(c)** (1 – 16.67%) [(600 x 1.10 x 10% - 175 x 8%)(1 – 0.40)] = 26 |
| 3. | **(d)** [(600 x 1.10 x 10% - 175 x 8%)(1 – 0.40)]/ 400 = 7.80% |
| 4. | **(b)** ROE(1 – d) = 7.80 x (1 – 0.1667) =6.50% |
| 5. | **(d)** $\frac{3.12(0.1667)(1.065)}{0.15- 0.065} = 6.51$ |

## ANSWERS TO CASE SCENARIO 45

| Question No. | Answer |
|---|---|
| 1. | **(d)** Money Market Hedging technique is used when derivatives to hedge the risk are not available. |
| 2. | **(b)** AUD 1000000/(1+0.05/4) = AUD 987654 |
| 3. | **(c)** AUD 987654 x ₹ 56.00 = ₹ 55308624 |

| | | |
|---|---|---|
| **4.** | **(d)** | ₹ 55308624(1 + 0.12/4) = ₹ 56967883 |
| **5.** | **(b)** | ₹ 56 (1 + 0.12/2)/(1 + 0.05/2) = ₹ 57.91 |

## ANSWERS TO CASE SCENARIO 46

| Question No. | Answer | |
|---|---|---|
| **1.** | **(b)** | ₹ 11,00,00,000 + ₹ 63,42,560 - ₹ 1,000 = ₹11,63,41,560 |
| **2.** | **(c)** | 1.64 = 95%, 1.96 = 98% and 2.58 = 99.50% |
| **3.** | **(b)** | 6 x 0.10 +7 x 0.25 +8 x 0.30 + 9 x 0.25 + 10 x 0.10 = 8% |
| **4.** | **(b)** | 4 x 0.10 + 6 x 0.20 + 8 x 0.40 + 10 x 0.20 + 12 x 0.10 = 8% |
| **5.** | **(a)** | $CV_X = \frac{1.14}{8} = 0.1425$ and $CV_Y = \frac{2.19}{8} = 0.274$ |

## ANSWERS TO CASE SCENARIO 47

| Question No. | Answer | |
|---|---|---|
| **1.** | **(a)** | ₹ 2,00,000 x 0.60 = ₹ 1,20,000 |
| **2.** | **(b)** | ₹ 7,00,000(1.10)(1.09) = ₹ 8,39,300 |
| **3.** | **(c)** | [₹ 6,00,000(1.10) – ₹ 3,00,000(1.12)]0.40 + ₹ 1,20,000 = ₹ 2,49,600 |
| **4.** | **(c)** | ₹ 4,00,000(1.12)(1.10) = ₹ 4,92,800 |
| **5.** | **(d)** | [{₹ 8,00,000(1.10)(1.09)(1.08) - ₹ 4,00,000(1.12)(1.10)(1.09)}0.40 + ₹ 1,20,000] PVF (10%, 3) |

## ANSWERS TO CASE SCENARIO 48

| Question No. | Answer | |
|---|---|---|
| **1.** | **(b)** | 22% X 0.70 + 24% X 0.30 = 22.60% |

| | | |
|---|---|---|
| **2.** | **(c)** | Variance = $(40)^2(0.70)^2 + (38)^2(0.30)^2 + 2(0.70)(0.30)(40)(38)(0.72) = 1373.61$, SD = $\sqrt{1375.61} = 37.08$ |
| **3.** | **(b)** | (0.72)(0.40)(0.38) = 0.1094 |
| **4.** | **(a)** | Solve simultaneous equations $22 = R_f + 0.86(R_m - R_f)$ and $24 = R_f + 1.24(R_m - R_f)$ |
| **5.** | **(d)** | Solve simultaneous equations $22 = R_f + 0.86(R_m - R_f)$ and $24 = R_f + 1.24(R_m - R_f)$ |

## ANSWERS TO CASE SCENARIO 49

| Question No. | Answer | |
|---|---|---|
| **1.** | **(d)** | (\$1.91 x 0.25 + \$1.95 x 0.60 + \$2.05 x 0.15) £ 300000 = £ 5,86,500 |
| **2.** | **(a)** | \$1.96 x £ 300000 = \$ 5,88,000 |
| **3.** | **(b)** | Borrow \$, convert to £, invest £, repay \$ loan in 180 days<br>Amount in £ to be invested = 3,00,000/1.045 = £ 2,87,081<br>Amount of \$ needed to convert into £ = 2,87,081 x 2 = \$ 5,74,162<br>Interest and principal on \$ loan after 180 days = \$5,74,162 x 1.055 = \$ 6,05,741 |
| **4.** | **(c)** | [(\$1.91 + 0.04) x 0.25 + (\$1.95 + 0.04) x 0.60 + (\$1.97 + 0.04) x 0.15] £ 3,00,000 + \$12,000 x 5.5% = \$5,95,560 |
| **5.** | **(c)** | \$1.91 x 0.25 + \$1.95 x 0.60 + \$2.05 x 0.15 = \$1.955 say \$1.96 |

## ANSWERS TO CASE SCENARIO 50

| Question No. | Answer | |
|---|---|---|
| **1.** | **(b)** | P Ltd. is planning to borrow after 6 months for a period of 3 months and higher of two rates shall be taken. |

| | | |
|---|---|---|
| **2.** | **(c)** | $₹\ 60\ \text{crore}\left[\frac{(9.60-9.30)}{1+\frac{0.096}{4}}\times\frac{1}{4}\right] = ₹\ 439{,}453$ |
| **3.** | **(a)** | $₹\ 60\ \text{crore}\left[\frac{(8.80-9.30)}{1+\frac{0.088}{4}}\times\frac{1}{4}\right] = ₹\ 733{,}855$ |
| **4.** | **(d)** | FRA is a traditional technique to manage interest rate risk. |

## ANSWERS TO CASE SCENARIO 51

| Question No. | Answer | |
|---|---|---|
| **1.** | **(d)** | Names of Harry Markowitz and William Sharpe are associated with Portfolio Theory and Black Scholes with Derivative Valuation. |
| **2.** | **(b)** | Refer the concept of Sustainable Growth Rate. |
| **3.** | **(b)** | Refer facts from case scenario. |
| **4.** | **(b)** | Refer the concept of Sustainable Growth Rate. |
| **5.** | **(c)** | Refer the concept of Sustainable Growth Rate. |

## ANSWERS TO CASE SCENARIO 52

| Question No. | Answer | |
|---|---|---|
| **1.** | **(c)** | The amount available for Interest Payment =Total PBT (₹ 136.79 Lakh) Less PBT of X Ltd. (₹ 114.29 Lakh) = ₹ 22.50 Lakh<br>Cash can be offered = ₹ 22.50 Lakhs/0.15 = ₹ 150 Lakhs |
| **2.** | **(b)** | EPS (Y Ltd.) = ₹ 8.5 & EPS (X Ltd.) = ₹ 4<br>Exchange Ratio = 8.5/4 = 2.125 |

| | | |
|---|---|---|
| **3.** | **(a)** | No. of Shares of Y Ltd.=1.8529 Lakh Shares [₹ 15.75 Lakh/ ₹ 8.50]<br>No. of shares to be issued by X Ltd. = 1.8529 Lakh * 2.125<br>= 3.9375 Lakh |

## ANSWERS TO CASE SCENARIO 53

| Question No. | Answer | |
|---|---|---|
| **1.** | **(a)** | Since option is lapsed. Thus, only loss is the loss of premium i.e., ₹ 25. |
| **2.** | **(b)** | Spot price is more than exercise price. Hence Call option.<br>Profit = ₹ 160 - ₹ 140 = ₹ 20<br>Net profit = ₹ 20 - ₹ 20 = ₹ 0 |
| **3.** | **(a)** | Spot price is Less than exercise price. Hence Put option.<br>Profit = ₹ 200 - ₹ 175 = ₹ 25<br>Net profit = ₹ 25 - ₹ 15 = ₹ 10 |

## ANSWERS TO CASE SCENARIO 54

| Question No. | Answer | |
|---|---|---|
| **1.** | **(b)** | Price of Bond B = 100*0.9091 + 100*0.8264 + 1100*0.7513 = ₹ 1000.<br>Duration of Bond B = [(90.91*1) + (82.64*2) + (826.43*3)] / 1000 = 2.74 |
| **2.** | **(a)** | Price of Bond A = 60*0.9091 + 60*0.8264 + 60*0.7513 + 60*0.6830 + 1060*0.6209 = ₹ 848.34<br>Duration of Bond A = [(54.55*1) + (49.58*2) + (45.08*3) + (40.98*4) + (658.15*5)] /848.34 = 4.41 (Approx.) |
| **3.** | **(c)** | Weighted average of duration |

| | |
|---|---|
| | <table><tr><td>Bond-A</td><td>4.41</td><td>75%</td><td>3.308</td></tr><tr><td>Bond-B</td><td>2.74</td><td>25%</td><td>0.685</td></tr><tr><td colspan="3">Average Duration</td><td>3.993</td></tr></table> Price Sensitivity = 3.993/1.10 = 3.63 |

## ANSWERS TO CASE SCENARIO 55

| Question No. | Answer | |
|---|---|---|
| **1.** | **(c)** | ß = 0.3733 × (18/12) = 0.560<br>Expected Return = 13% + 0.560 (20%-13%) = 16.92% |
| **2.** | **(a)** | PV of dividend= 6.25*0.855 + 7.81*0.732 + 9.77*0.626 + 12.21*0.535 + 15.26*0.458 + 19.075*0.391 = ₹ 38.16<br>PV of Terminal Value = [(19.075(1.10)/(0.1692-0.10)] × 0.391 = ₹ 118.53 |
| **3.** | **(a)** | Since current Market Price per share of the share is ₹ 315 higher than Intrinsic Value of the share, the stock is overvalued. |

## ANSWERS TO CASE SCENARIO 56

| Question No. | Answer | |
|---|---|---|
| **1.** | **(c)** | MPS = ₹ 200 × 15 = ₹ 3000 per share<br>Original shares = (₹ 3000 Crore)/( ₹ 3000) = 1 crores shares<br>No. of shares proposed to be bought back = 1 crores × 20% = 20 Lakhs shares |
| **2.** | **(a)** | Post buy back EPS = (₹ 3057 per share)/15 =₹ 203.80 per share<br>Post buy back Earnings after tax = ₹ 203.80 × 80 Lakhs shares<br>= ₹ 16,304 Lakhs<br>Post buy back Earnings before tax (A) = (16,304)/(1 - 0.30)<br>= ₹ 23,291. 43 Lakhs<br>Pre buy back Earnings after tax = 200 × 100 Lakhs shares<br>= ₹ 20,000 Lakhs |

| | | |
|---|---|---|
| | | Pre buy back Earnings before tax (B) = (20,000)/(1 - 0.30)<br>= ₹ 28,571.43 Lakhs<br>Interest to be paid for availing bank loan (A) - (B) = ₹ 5,280 Lakhs |
| **3.** | **(c)** | Loan amount to be raised = ₹ 5,280 Lakhs /0.08 = ₹ 66,000 Lakhs |
| **4.** | **(c)** | Buy back price per share = ₹ 66,000 Lakhs /20 Lakhs = ₹ 3,300 per share<br>Premium per share paid over the current MPS = ₹ 3300 - ₹ 3000 = ₹ 300 per share |
| **5.** | **(c)** | % Premium = ₹ 300/ ₹ 3000 × 100 = 10% |